# ■ PulmoTwin

## A Multimodal Cardiopulmonary Digital Twin for Proactive Adverse Event Forecasting & Personalized Clinical Intervention

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| **Competition:** | Digital Twin Challenge 2026 |
| **Organizer:** | Happiest Health — Reimagining & Reforming Healthcare in India Summit 2026, Bengaluru |
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| **Repository:** | https://github.com/Kabirroy12345/Digital_Twin.git |
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# 1. The Healthcare Challenge: India's Chronic Respiratory Burden

*Why Reactive Asthma and COPD Care Fails Patients in Indian Metros*

| ■ 37.9M Asthmatics in India | ■ The Reactive Care Trap | ■ Clinical Memory Void |
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| India accounts for over 42% of global asthma deaths despite housing only 13% of asthmatic patients. Seasonal smog spikes, temperature inversions, and high PM2.5 concentrations trigger recurring acute respiratory emergencies. | Patients only react *after* bronchospasm occurs. By the time chest tightness and wheezing set in, airway resistance has already spiked by 300-500%, requiring emergency nebulization or hospitalization. | Under GINA guidelines, using reliever inhalers >2x/week indicates uncontrolled asthma. Yet over 80% of patients fail to accurately recall 30-day symptom frequency and puff counts during brief doctor consultations. |

# 2. Introducing PulmoTwin: The Cardiopulmonary Digital Twin

*Dynamic Virtual Replica of Patient Physiology Synchronized with Real-Time Streams*

| Component | Physiological / Technical Function | Real-World Data Source |
|---|---|---|
| **Virtual Airway Dynamics** | Models airway resistance (Raw), dynamic lung compliance (Cdyn), and bronchial hyperresponsiveness. | Derived from baseline spirometry (FEV1, FVC, personal best PEF) & smart spirometers. |
| **Autonomic & Cardio Twin** | Models sympathetic tone vs parasympathetic vagal activity; detects autonomic distress before dyspnea. | Continuous PPG wearables (Heart Rate, HRV/rMSSD, Pulse Oximetry SpO2). |
| **Environmental Integration** | Projects hourly particulate and chemical irritant load on respiratory mucosal membrane. | Live Open-Meteo Air Quality & Weather API (PM2.5, NO2, SO2, O3, Temp, Humidity). |
| **Predictive Event Horizon** | Forecasts acute exacerbation probability 4h, 12h, and 24h into the future using differential kinetics. | Multimodal collaborative ensemble + differential physiological modeling. |

# 3. Counterfactual 'What-If' Simulation Sandbox

*Optimizing Personalized Treatment Protocols in Virtual Reality Before Physical Execution*

| Scenario Tested | Projected Airway Raw | Peak Exacerbation Risk (ERI) | Clinical Outcome |
|---|---|---|---|
| **Baseline (No Proactive Action)** | Surges to 8.4 cmH2O/(L/s) | 0.84 (RED - Critical) | Severe nocturnal bronchospasm during 20:00 smog peak. |
| **Scenario A: Indoor HEPA Air Purifier** | Restricted to 4.2 cmH2O/(L/s) | 0.46 (YELLOW - Caution) | 45.2% risk reduction; eliminates indoor particulate irritant load. |
| **Scenario B: Proactive SMART Inhaler** | Maintained at 2.6 cmH2O/(L/s) | 0.28 (GREEN - Controlled) | 66.7% risk reduction; bronchodilator pre-empts airway constriction. |
| **Scenario C: Combined Proactive Protocol** | Optimal: 2.1 cmH2O/(L/s) | 0.19 (GREEN - Low Risk) | 77.4% risk reduction; complete prevention of preventable exacerbation. |

# 4. Clinical Safety Rails & Human-in-the-Loop Governance

*Deterministic Guardrails Guaranteeing Patient Safety with Zero LLM Hallucination Risk*

| ■■ Deterministic Input Interceptor | ■ Output Sanitizer Filter | ■ Zero-Trust Approval Queue |
| --- | --- | --- |
| Instant regex and keyword scanning detects acute respiratory distress in English and Hindi ('saans nahi aa rahi', 'blue lips', 'silent chest'). Immediately bypasses the LLM to trigger emergency 108/112 ambulance protocols with zero API token latency. | Post-generation safety filters scrub all model outputs. Completely blocks unauthorized prescription changes, dosage alterations, or diagnostic assertions, enforcing adherence to GINA Step-5 physician plans. | External actions (SMS caregiver alerts, n8n webhook notifications, calendar adjustments) are strictly gated behind explicit human authorization. The agent proposes; the patient retains sovereign authority. |

# 5. Conclusion & The Summit Vision

*Reimagining & Reforming Healthcare in India Summit 2026, Bengaluru*

| Key Project Milestone | Achievement & Alignment with Challenge |
|---|---|
| **Full Working Prototype** | Flask backend + Interactive Editorial Web UI + Real-time Open-Meteo sync + SQLite audit trail. |
| **Wearable + EHR Architecture** | Continuous PPG telemetry stream + ABDM / FHIR R4 schema integration. |
| **Forward Predictive Simulation** | 24-hour forward physiological trajectory forecasting acute exacerbations before onset. |
| **Counterfactual Intervention Sandbox** | Empowers patients and clinicians to simulate the clinical efficacy of interventions virtually. |
| **Clinical Consultation PDF Generator** | Instantly synthesizes 30-day longitudinal GINA clinical reports for physician appointments. |
| **Bengaluru Summit 2026 Showcase** | Ready for live jury presentation, demonstration, and clinical translation with Happiest Health. |